

KALEVALA, Russian Federation

WMO: 224080

Lat: **65.217N**Lon: **31.150E**Elev: **394**StdP: **14.49**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-20.5	-31.3	1.0	-27.0	-24.4	1.4	-20.5	16.8	21.0	14.8	23.0	0.9	340	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	78.0	64.4	74.4	62.6	71.2	60.8	66.2	75.2	64.2	72.1	62.1	69.4	5.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	87.2	69.8	60.8	80.6	68.6	58.6	74.5	66.3	31.1	75.4	29.6	72.1	28.0	69.6	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.3	13.2	11.4	-32.2	81.8	7.3	3.8	-37.5	84.5	-41.8	86.7	-45.9	88.8	-51.2	91.6		
(5)			-32.3	68.7	7.3	3.1	-37.6	70.9	-41.8	72.7	-45.9	74.4	-51.2	76.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.6	11.3	11.6	20.6	32.0	43.4	54.1	60.4	55.6	46.6	35.8	24.9	17.1	(6)
(7)		DBStd	19.57	15.08	15.03	10.44	8.11	8.43	7.22	6.34	6.15	6.40	7.42	10.32	14.00	(7)
(8)		HDD50	6378	1199	1076	912	541	239	38	4	16	140	440	752	1021	(8)
(9)		HDD65	11143	1664	1496	1377	991	669	335	169	298	552	904	1202	1486	(9)
(10)		CDD50	750	0	0	0	0	36	160	327	189	37	1	0	0	(10)
(11)		CDD65	40	0	0	0	0	0	7	27	6	0	0	0	0	(11)
(12)		CDH74	333	0	0	0	0	14	80	189	50	0	0	0	0	(12)
(13)		CDH80	40	0	0	0	0	1	8	26	5	0	0	0	0	(13)
(14)	Wind	WSAvg	4.5	4.2	4.4	5.1	4.6	4.9	4.7	3.9	3.7	4.3	4.9	4.7	4.6	(14)
(15)	Precipitation	PrecAvg	20.9	1.2	1.0	0.9	1.1	1.5	2.4	2.7	2.9	2.5	1.8	1.8	1.2	(15)
(16)		PrecMax	28.3	2.8	2.6	2.1	3.3	3.0	4.3	7.8	4.8	4.1	3.5	3.5	2.1	(16)
(17)		PrecMin	16.3	0.3	0.2	0.1	0.1	0.2	1.1	0.7	1.1	1.0	0.5	0.1	0.3	(17)
(18)		PrecStd	2.9	0.6	0.6	0.5	0.7	0.7	1.0	1.4	1.0	0.7	0.8	0.7	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	38.8	44.5	59.0	75.9	80.9	83.4	79.2	67.6	54.3	46.2	39.1	(19)
(20)			MCWB	35.1	35.4	37.5	48.7	58.9	64.8	67.5	66.1	58.2	50.9	43.6	37.0	(20)
(21)		2%	DB	34.0	35.1	39.9	51.3	69.2	75.9	79.0	74.5	63.1	50.5	42.2	36.3	(21)
(22)			MCWB	32.9	33.4	34.7	43.0	55.5	62.5	65.6	63.5	55.4	48.1	40.8	35.0	(22)
(23)		5%	DB	32.3	33.1	36.6	47.3	63.2	71.5	75.5	70.6	59.7	48.5	38.8	34.5	(23)
(24)			MCWB	31.6	31.9	33.1	39.9	51.6	59.9	64.1	61.3	53.8	46.6	37.5	33.5	(24)
(25)		10%	DB	30.0	30.8	34.1	43.7	58.6	67.4	72.3	67.2	57.0	45.9	36.2	32.8	(25)
(26)			MCWB	29.2	29.5	31.7	37.8	49.0	57.4	61.9	59.4	52.7	44.1	35.3	31.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.4	35.9	38.4	48.1	61.1	66.9	70.6	67.8	59.6	51.3	44.3	37.4	(27)
(28)			MCDB	36.9	37.7	43.0	56.8	74.0	77.5	79.3	75.5	65.4	53.0	45.2	38.0	(28)
(29)		2%	WB	33.1	33.8	35.5	43.7	56.6	64.2	67.2	64.5	56.8	48.9	40.9	35.2	(29)
(30)			MCDB	33.8	35.1	38.6	50.4	66.9	73.7	75.8	72.6	61.2	50.2	42.2	36.0	(30)
(31)		5%	WB	31.7	32.1	33.6	40.6	52.8	61.5	65.3	62.3	54.9	46.6	37.8	33.5	(31)
(32)			MCDB	32.3	33.1	35.8	46.1	61.8	69.8	73.2	69.0	58.6	48.1	38.7	34.3	(32)
(33)		10%	WB	29.2	29.6	32.0	38.3	49.8	58.1	63.3	60.1	53.1	44.2	35.5	32.0	(33)
(34)			MCDB	30.0	30.8	34.0	42.9	58.0	65.7	70.4	65.9	56.4	45.8	36.1	32.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.2	14.1	16.4	15.9	16.5	16.6	16.0	15.9	13.1	8.1	8.7	11.0	(35)
(36)			MCDBR	9.9	10.4	15.5	21.4	25.6	23.1	21.1	21.8	18.1	9.5	7.4	7.6	(36)
(37)			MCWBR	9.8	9.8	12.4	13.9	14.7	12.1	10.4	12.2	11.5	8.1	7.1	7.2	(37)
(38)		5% WB	MCDBR	9.5	10.2	12.5	19.4	23.6	20.6	18.7	19.0	15.7	9.0	6.9	7.4	(38)
(39)			MCWBR	9.5	9.8	10.4	12.9	14.4	11.8	10.3	11.3	10.8	8.1	6.8	7.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.204	0.272	0.303	0.336	0.347	0.343	0.368	0.344	0.320	0.295	0.352	0.202	(40)	
(41)		taud	1.932	2.100	2.162	2.292	2.448	2.488	2.442	2.482	2.529	2.374	2.640	1.957	(41)	
(42)		Ebn at Noon	90	190	241	263	273	277	265	261	243	188	88	22	(42)	
(43)		Edh at Noon	7	19	28	32	30	30	30	26	20	15	7	2	(43)	
(44)	All-Sky Solar Radiation	RadAvg	38	214	703	1226	1459	1625	1515	1111	619	215	49	10	(44)	
(45)		RadStd	3	41	64	101	156	110	163	104	68	30	8	1	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.10	N/A	N/A	N/A	N/A	N/A	N/A	-436	N/A	N/A
(47) Regional (16 neighbors)		+1.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page